

NITDGP/Btech/Reg/ODD/2025-26

NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR
B.Tech 3rd Sem; Odd Semester END-Term Examination, 2025-26

Course Code: ESC 331

Course Name: Engineering Geology for Civil Engineering

Full Marks: 60

Time: 180 Minutes

Question No.	Body of the Question	Marks
1	a) Define strike and dip of a planar structure. Draw a block diagram accordingly. b) What would be the strike and dip of a horizontal plane? c) Define tension joint and compression joint in structural geology. d) Discuss the positive effects of joint structures.	3 + 2 + 2 + 3 = 10
2	a) Define hinge point and inflection point of a fold structure. b) Define interlimb angle of a fold structure. Also give a classification of fold based on interlimb angle. c) Write a note on shattering of rocks in fold structure.	2 + 5 + 3 = 10
3	a) Differentiate fault zone from shear zone in geology. b) Define hanging wall block and footwall block of a fault structure. c) Define normal fault and reverse fault. d) Write a note on effects of faulting on layered rocks.	2 + 2 + 2 + 4 = 10
4	a) Differentiate dead-storage water surface from minimum water surface in the reservoir. b) Define sluice in a dam. Also write the purposes of sluice. c) Write a note on problems occur during and after the construction of dam on faulted rocks.	2 + 3 + 5 = 10
5	a) What is uplift pressure in a dam? b) How does uplift pressure occur? c) Write on consequences of uplift pressure. d) How to mitigate the problems related with uplift pressure?	1 + 3 + 2 + 4 = 10
6	a) Define overbreak in tunnel. Also mention the chief factors which cause overbreak. b) What does 'pay line' in tunneling mean? c) Write a very short note on tunneling through inclined strata/layers. d) Name the basic types of tunnel construction in common use.	3 + 1 + 3 + 3 = 10
7	a) Define the 'modulus of elasticity' of materials. Name the types of rock based on this property. Also state the stress-strain relationship of these rocks when load is applied. b) Define building stone. Name the most widely used rock as building stone. Name a rock which commonly use as roofing material in village area. c) 'Texture and structure of the rocks also play an important role in choosing better building stone' - explain it.	4 + 3 + 3 = 10
8	a) How does expansive soil affect landslides? b) Define sensitive soil. c) How does gradient of the slope influence the slope stability? Also discuss the role of water in slope instability (downslope movement of earth materials).	3 + 1 + 6 = 10
9	a) Discuss different ways of materials (sediments) transport by stream/river. b) Write a short note on chief drainage pattern.	5 + 5 = 10